Luvian

The Context Layer for Engineering Intelligence

Built for an autonomous future.

Seed Round • Confidential • April 2026

THE PROBLEM

Enterprise AI keeps failing. The model isn't the reason.

Your AI already has access to the data it needs. The problem is the data itself: disconnected, conflicting, stale, permission-fragmented, and semantically weak. That isn't a model problem. It's a context problem.

95% of enterprise GenAI pilots produce zero ROI — MIT NANDA, Jul 2025

$30–40B already invested

40%+ of agentic AI projects will be cancelled by 2027 — Gartner, Jun 2025

Fragmented Context

Requirements live in one tool, decisions in chat, code in another, tests in a third, telemetry in a data lake. None of it speaks the same language.

No System Semantics

LLMs see strings of text. They don't see joints, sensors, control loops, interfaces, or safety constraints. Without typed engineering meaning, reasoning hallucinates.

No Persistent Memory

When an engineer leaves, the rationale walks out with them. The same decisions get re-litigated. Tribal knowledge quietly evaporates from one release to the next.

MARKET OPPORTUNITY

~$70B market.
Growing 11–14% CAGR (mid-case).

TAM (2025)

$58–95B

Engineering intelligence ceiling: PLM + ALM + MBSE + industrial AI + KG + A&D digital engineering

SAM (2027)

$6–12B

Addressable to a typed, versioned, permission-aware engineering graph

SOM (YEAR 3)

$20–80M ARR

15–40 strategic accounts in A&D primes, defense tech, tier-1 auto/robotics

$26–37B
PLM (6.6–8.7% CAGR)

$6–25B
Industrial AI (17.9–19.5%)

$6.3B
A&D Digital Engineering (11.6%)

$3.5–5.6B
ALM / Requirements (7.4–10.7%)

$2.9B
Enterprise KG (21.3% CAGR)

Why now: **DoD MOSA mandate (Dec 2024)** forces model-based engineering • **Robotics raised $13.8B in 2025**, up from $7.8B • **SysML v2 ratified** — typed engineering data is now standard

Sources: Research and Markets, Mordor, Dataintelo, MarketGenics, 360iResearch, Grand View, DoD Comptroller. Full segment buildup in

THE SOLUTION

The context layer for engineering intelligence.

Luvian connects requirements, designs, code, tests, telemetry, and decisions into one continuously evolving knowledge graph — so AI reasoning, traceability, and engineering automation can finally be trusted.

Context Acquisition

Connectors into every system where engineering knowledge already lives — modelers, trackers, source, CI, PLM, test, simulation, telemetry.

Semantic Normalization

A canonical engineering ontology with stable identity, lineage, ownership, and temporal state. Typed system meaning, not strings.

Context Orchestration

Beyond RAG. Permission-aware retrieval, graph traversal, temporal reasoning, contradiction detection, and confidence scoring.

AI Reasoning

Copilots, reviewers, and domain agents that earn trust because the substrate underneath them does.

PRODUCT

Shipping today. Building the stack.

We started with the structured-artifact layer: a SysML v2-native modeler. It's the seed of the broader knowledge graph — a typed, collaborative, AI-assisted source of truth that the rest of the platform federates around.

AVAILABLE NOW

- ✓ **SysML v2 Structured Artifact Layer** — blocks, ports, interfaces, connections. Every element feeds the graph as a typed semantic entity.
- ✓ **Real-time Collaborative Modeling** — Yjs/CRDT multi-user editing with presence, awareness, and operation-level undo.
- ✓ **Requirements & Safety Workflows** — INCOSE quality, traceability, HARA/TARA/STPA/SOTIF/FMEA, V&V with evidence attachment.
- ✓ **AI-assisted Editing** — block, port, and requirement suggestions grounded in the graph.
- ✓ **Open Export** — SysML v2 JSON, ReqIF, ontology-aware. Your data, your format, no lock-in.

COMING SOON

- **Engineering Connectors** — issue trackers, requirements platforms, source control, CI/CD, PLM, test mgmt, simulation, telemetry, all pulled into one canonical graph.
- **Engineering Ontology** — canonical types with stable identity, lineage, temporal state, and ownership across every artifact.
- **Context Orchestration Layer** — permission-aware retrieval, graph reasoning, contradiction detection, and confidence scoring beyond RAG.
- **Domain Reasoning Agents** — safety review, requirement quality, impact analysis, test synthesis, and compliance agents grounded in your context.
- **Governance & Trust** — provenance, confidence scoring, audit trail, and contradiction surfacing so AI output is reviewable, not just persuasive.

WHY LUVIAN WINS

Technical moat.

Semantic Engineering Graph

A typed, versioned, permission-aware graph of every artifact, link, and decision across the product lifecycle. The substrate enterprise AI needs to be trustworthy in engineering.

Patent-pending

Beyond-RAG Context Orchestration

Permission-aware retrieval, graph traversal, temporal reasoning, contradiction detection, and confidence scoring. Reasoning over connected context, not just chunked retrieval.

Patent-pending

Persistent Organizational Memory

Decisions, rationale, and trade-offs become machine-readable and continuously evolving. People leave; the memory stays. Tribal knowledge becomes operational graph.

Audit-ready provenance

Air-Gap Ready Local Reasoning

Commercial (cloud LLM) and government (local-only) profiles from a single codebase. The context graph and reasoning agents can run entirely behind your firewall — your engineering data never leaves.

Single codebase, dual profiles

BUSINESS MODEL

SaaS + on-prem. 4 tiers.

Tier	Target	Price / seat / yr	Key Includes
Team	Small teams (5–20)	**$1,200** ($100/mo)	Modeling, requirements, real-time collab — your foundation in the graph
Professional	Mid-market (20–100)	**$2,400** ($200/mo)	+ Safety analysis, requirement quality, AI copilots, public API
Enterprise	Large programs (100+)	**$3,600** ($300/mo)	+ Connectors, on-prem, SSO/SAML, audit logs, SLA, dedicated support
Defense/Gov	DoD, air-gap (50+)	**$5,400** ($450/mo)	+ Air-gap reasoning, IL4/IL5 docs, FedRAMP artifacts, dedicated CSM

70–75%
Subscription Revenue

15–20%
Professional Services

5–8%
AI Compute Add-On

2–5%
Platform / API

3–10x cheaper than incumbent MBSE suites (Cameo, DOORS, Jama). Premium positioning vs. point-tool AI plays — differentiation is the connected engineering context graph underneath.

GO-TO-MARKET

Land in robotics. Expand across the lifecycle.

Y1

Design Partners

Founder-led sales. 5 partners. Subsidized pricing for feedback + case studies.

Y2

Early Revenue

First AE. Inbound from content + community. 19 customers. SOC 2.

Y3

Growth

Sales team. Channel partners (Booz Allen, SAIC). FedRAMP. 46 customers.

Y4-5

Scale

Full sales org. Defense enterprise. International. 131 customers.

Target Verticals — Built for the people building robots

- Industrial Robotics
- Cobots
- Warehouse Automation
- Mobile Robotics & AMRs
- Drones & UAS (defense + commercial)
- Mechatronics & SDV stacks

Anywhere firmware, mechanical, perception, control, and safety have to integrate cleanly — and anywhere AI without trustworthy context is dangerous.

COMPETITIVE LANDSCAPE

Connected graph + local-first reasoning.

Horizontal Reach × Vertical Depth

Capability Coverage

Capability	Luvian	Flow	Trace	Aras	Cameo	Glean	Palantir
Typed Engineering Ontology	✓	~	~	~	✓	×	~
SysML v2 (post-Dec 2025)	✓	×	×	×	✓	×	×
Requirements + Trace	✓	✓	✓	✓	~	×	~
Safety Methods (HARA/TARA/STPA)	✓	×	×	~	×	×	×
V&V / Verification Evidence	✓	~	~	~	×	×	~
Beyond-RAG Reasoning	✓	~	~	~	×	~	✓
True Air-Gap (IL5/IL6)	✓	×	~	✓	✓	~	✓

Glean shipped on-prem via Dell (May 2025); Cameo shipped SysML v2 in 2026x (Dec 2025) — both partially close historical Luvian gaps. The remaining wedge: typed engineering ontology + safety methods + true offline air-gap, in one graph.

Only Luvian unifies SysML v2 modeling, safety methodology engines, requirements/V&V traceability, and beyond-RAG reasoning in one typed engineering graph — deployable on-prem and air-gapped. Detailed competitor profiles in market-eval-v2.

DEFENSE & GOVERNMENT

Built for classified. Deployable anywhere.

Air-Gap Ready Reasoning

Single `docker compose up` deploys the full context layer. Local-only embeddings (MiniLM, 80MB) and local LLM-backed reasoning. Zero data egress.

Docker · ONNX Runtime · Local LLM

Engineering Standards Literacy

DO-178C, DO-254, ARP4754A/4761, ISO 26262, ISO 21434 wired into the ontology, safety analysis, and V&V workflows.

Avionics · Automotive · Defense

Compliance & Provenance

SOC 2 Type 1 post-seed. FedRAMP Li-SaaS Y3. IL4/IL5 path Y4. Classification-aware permissions, provenance, and audit trail on every graph edge.

SOC 2 · FedRAMP · IL4/IL5

$270K+	$5M+	$15M+
Min defense ACV (50 seats)	Enterprise-wide deal potential	Max defense ACV (500+ seats)

TEAM

Built by engineers who live in the domain.

F

Founder & CEO

Full-stack architect. Built the 3-layer rendering pipeline, ambient ML engine, and defense deployment architecture. Domain expertise in systems engineering tooling.

SW

SW Engineer

20-year veteran. Production hardening: CI/CD, security, observability, performance. Enterprise infrastructure at scale.

PM

Product Manager

Product launch and market fit. Personas, GTM strategy, analytics, onboarding, and beta program execution.

Post-Funding Hiring Plan (Y1)

- Head of Sales
- Solutions Engineer
- Senior Frontend
- Backend / ML Engineer
- Ops / G&A

FINANCIAL PROJECTIONS

Path to $21M revenue by Year 5.

Revenue & Costs (Base Case)

Revenue Scenarios

Base Case	Y1	Y2	Y3	Y4	Y5
Revenue	$165K	$846K	$3.3M	$9.9M	**$21.2M**
Costs	$1.7M	$3.7M	$7.2M	$11.8M	$16.8M
Net Income	($1.6M)	($2.9M)	($3.9M)	($1.9M)	**$4.4M**

UNIT ECONOMICS

Best-in-class LTV:CAC.

LTV:CAC Ratio by Segment (target: >3x)

Gross Margin (blended) **75–80%**

NRR (Y5 target) **130%**

CAC Payback **3–5 mo**

Rev / Employee (Y5) **$299K**

Benchmark: $300–500K for best-in-class SaaS

THE ASK

Raising $7–10M Seed

24+ months runway to context layer GA and first $1M ARR.

Build the Team

4 eng (graph + ontology + connectors), 1 product, 1 SE, 1 sales, 1 ops. Hire the team that scales past the founder.

Close First Revenue

5 design partners in robotics & mechatronics. SOC 2 Type 1. First enterprise contracts. $100K+ ARR by month 12.

Establish the Category

Define "engineering intelligence" before incumbents respond. File patents on context orchestration and semantic engineering graph. Build advisory board.

Funding Milestones

Seed (now)	Series A (Y2)	Series B (Y3–4)
$7–10M	**$20–30M**	**$50–80M**
PMF signal: 3+ paying, NPS > 40	$1M+ ARR, 120%+ NRR	$5M+ ARR, defense contracts

Luvian

The context layer for engineering intelligence.

Built for an autonomous future.

Market	~$70B TAM ($58–95B range), $6–12B SAM
Model	$100–$450/seat/mo SaaS + on-prem
Raise	$7–10M seed, 24+ months runway
Moat	Semantic engineering graph + beyond-RAG reasoning, air-gap ready

Luvian Confidential — NDA Restrictions